实验部分严格审稿与融合重写方案

针对 Section XI: Numerical Experiments

基于 `bare_jrnl_new_sample4(7).pdf`

本报告从严格审稿人的角度检查实验部分的逻辑、证据链、创新点呈现、巡检应用背景、基线公平性、统计规范和版面组织。重点回答两个问题：**现有六组算例是否过多，以及哪些实验可以融合而不损失证据强度。**

审稿结论：**Major Revision**

建议保留现有数据证据，但将六个 case studies 重组为四个研究问题，并补充少量关键对照。

目录

1 总体审稿结论

当前实验覆盖了组件消融、首解质量、anytime 优化、规模扩展、基线比较和机器人失效修复，证据类型本身是充分的。问题不在于“实验数量一定过多”，而在于：**每一个算法模块被单独写成一个 case study，导致同一贡献被拆成多段、相同解释反复出现、不同实验使用不同公式和机器人配置，读者难以形成统一证据链。**

一句话结论

六组实验不需要机械删除，推荐将其融合为四个研究问题：

1. integrated automaton construction 的组件消融；
2. first-solution quality 与 warm-start BnB 的 anytime 性能；
3. 规模扩展与外部基线比较；
4. robot unavailability 下的在线修复。

这样可以保留现有数据，同时显著压缩重复文字，并使四组证据分别对应论文四个核心主张。

从审稿角度，现有实验有以下五个最需要优先修复的问题。

等级	当前问题	审稿人可能提出的质疑	建议处理
Major	“Scalability”实验同时改变机器人数量和任务需求	不能区分计算量增加究竟来自 robot pool 还是 collaboration demand	固定任务需求后改变机器人数量；或将现有实验改名为 coupled stress test
Major	当前消融不是独立因素设计	Variant II 同时关闭 GBA pruning 和 on-the-fly planning，无法单独归因	改为 2×2 因子消融，或至少增加“无早期剪枝但保留 on-the-fly”
Major	Table III 的 plan-graph best 与 global optimum 不一致	与全文可能暗示的原问题全局最优性冲突	将 global optimum 改为 reference optimum；明确 BnB 只优化 retained plan graph
Major	在线修复没有从头重规划对照	绝对毫秒数不能证明 retained graph 的贡献	增加 full replanning from current state 作为直接基线
Major	Section XII 为空，但摘要和结论声称有 physical experiments	属于明显的证据缺失与前后矛盾	补充物理实验，或删除所有相关声明

2 六组实验是否太多：推荐的融合结构

2.1 推荐的四组证据链

表 2: 现有六组实验的融合建议

现有内容	推荐新位置	融合理由	主要输出
A. Warm-Started BnB C. First-Solution Quality	New B. Anytime Plan Quality and Optimization	两者回答同一问题：早期获得的计划是否可用，以及后续 BnB 能否快速改进。Fig. 2 与 Table III 应放在同一小节。	J_{first}、J_{PG}、time-to-best、pruning ratio、fixed-budget gap
B. Planning-Aware Construction	New A. Component Ablation	Tables I–II 都在验证 GBA early pruning 与 on-the-fly planning，应合并为一张两面板消融表	T_{first}、NBA transitions、plan-graph size、memory
D. Scalability E. Comparison With Baselines	New C. Scalability and Baseline Comparison	规模趋势和外部基线是同一层级的算法能力验证；共用 benchmark suite 和 team-size axis 后更紧密。	T_{first}、T_{best}、timeout rate、cost gap、speedup
F. Online Repair	New D. Online Repair Under Robot Unavailability	这是执行期鲁棒性证据，应保留独立小节，但增加从头重规划对照	repair latency、reassigned tasks、mission completion、success rate

2.2 推荐的最终 Section XI 结构

1. **XI-A Experimental Setting and Inspection Missions**
   只介绍一次工作空间、机器人类型、巡检任务族、统计协议和指标，不再在每个小节重复。
2. **XI-B Ablation of Integrated Automaton Construction**
   证明早期 GBA pruning 与 on-the-fly allocation 分别有效。
3. **XI-C Anytime Plan Quality and Warm-Started Optimization**
   合并当前 A 和 C，回答“早期计划是否可执行、是否足够好、后续是否可持续改进”。
4. **XI-D Scalability and Comparison With Baselines**
   合并当前 D 和 E，统一公式、随机种子、team-size axis 与 timeout。
5. **XI-E Online Repair Under Robot Unavailability**
   保留当前 F，增加 retained graph 与 full replanning 的对照。

严格来说，这是“一个设置小节 + 四组实验”，比当前“一个总述 + 六个算例”更紧凑，也更符合顶刊常见的 research-question-driven 组织方式。

3 与巡检背景的统一

3.1 当前背景存在的问题

论文前文已经提到 smart factory、warehouse、inspection、maintenance 和 monitoring，但数值实验仍主要以 a_i、l_i 和 type-j 的抽象编号呈现。严格审稿人会认为：**应用背景只停留在 Introduction，而没有进入 benchmark design、metric interpretation 和 failure scenario。**

例如，当前 Section XI 中多次列出

$$c(a_i) = (\tau_i, n_i, l_i),$$

但没有解释该 action 在巡检任务中表示什么、为什么需要多机器人同步、为什么首解时延在巡检中重要、机器人失效会导致什么实际风险。因此，实验虽然数据很多，却没有充分证明实际应用价值。

3.2 推荐的统一应用场景

建议将全文统一为：**大型工业设施的异构机器人巡检与维护协同。**该场景可以自然解释顺序、并行、同步、禁入和机器人失效等所有 LTL 特征。

表 3: 建议的巡检场景语义映射

元素	建议语义	在实验中的作用
$\mathcal{R}_1$	轮式视觉巡检机器人	大范围例行巡视、仪表读数、走廊检查
$\mathcal{R}_2$	空中或热成像巡检机器人	高处设备、热异常、难以到达区域检查
$\mathcal{R}_3$	气体/温度传感机器人	泄漏检测和危险区域确认
$\mathcal{R}_4$	操作与维护机器人	阀门操作、简单维修、故障隔离
$\mathcal{R}_5$	物资与工具运输机器人	将维修工具或备件送至目标区域
l_1–l_8	变电间、管线区、生产线、阀站、危化区、泵房、巡检走廊、维护/充电站	形成具有空间分离、障碍和禁入区域的工业设施
a_i	视觉检查、热检查、气体检查、协同复核、工具配送、阀门操作、维修确认等	对应不同机器人类型、数量和目标区域的 action requirement

不建议在正文逐一给 a_1–a_{11} 强行赋予完全不同的业务名称。更好的做法是在实验设置中给出一张 compact benchmark table，将四类公式解释为四类巡检任务：

1. **Routine multi-zone inspection**: 多个区域最终都要完成检查；
2. **Sequential inspection**: 某些检查必须在前置安全确认后开始；
3. **Safety-constrained inspection**: 检查期间必须避开危险区域或保持禁止条件；
4. **Inspection-to-maintenance mission**: 检测到异常后，在多个维修方案中选择并执行协同维护。

这样保留数学抽象，又能让读者理解公式和实验指标的实际意义。

4 逐小节严格审查与修改

4.1 Section XI 开头与 Experimental Setting

4.1.1 主要问题

- 开头用六个并列分号罗列 case studies，句子过长，且没有形成“问题–实验–结论”的层次。
- “This case study examines...”在 A–F 中反复出现，属于结构性冗余。
- 工作空间随机障碍、机器人初始位置随机，但没有统一说明重复次数、随机种子和统计方式。

- 文中只在 First-Solution Quality 处说明每个 case 运行 25 次，其余表格像单次结果，统计口径不一致。
- $T_{\text{search}} = \infty$ 不规范。应写为 TO，并明确 timeout 值。
- “time when the best solution is obtained” 中 best 的范围不明确，应区分 best incumbent、plan-graph optimum 和 reference optimum。

4.1.2 推荐改写

可直接替换的英文开头

This section evaluates the proposed method in a large industrial inspection scenario. The numerical study addresses four questions. First, we isolate the effects of GBA-level pruning and on-the-fly task allocation during NBA generation. Second, we evaluate the quality of the first executable plan and the subsequent improvement obtained by the warm-started branch-and-bound search. Third, we examine scalability and compare the proposed method with poset- and MILP-based baselines under common inspection missions and robot-team configurations. Finally, we evaluate online repair after robots become unavailable during execution.

All methods were implemented in Cython 0.29.24 and executed on a workstation equipped with a 3.40-GHz Intel Core i7-11390H CPU and 16 GB RAM. The workspace is a $40\,\text{m} \times 40\,\text{m}$ industrial facility containing eight inspection regions and randomly generated obstacles. Unless otherwise stated, all methods use the same workspace, robot initial states, and random seeds. We report the time to the first feasible plan, T_{first}, the time at which the best incumbent is obtained, T_{best}, and the termination time, T_{search}. A run that does not terminate within the prescribed time limit is marked TO. For randomized experiments, we report the mean and standard deviation over repeated trials.

4.2 当前 A 与 C：Warm-Start 和 First-Solution Quality 应融合

4.2.1 当前证据的优点

当前 A 用一个代表性运行展示 BnB 改进过程，当前 C 用四个公式展示 first solution 与 plan-graph best 的关系。两者互补：前者说明动态过程，后者说明跨任务稳定性。因此它们不应分开写。

基于现有数据，可以得到以下清晰结论：

- 代表性算例中，$J_{\text{first}} = 47.21$ 降至 $J_{\text{best}} = 43.32$，相对降低约 8.24%；
- BnB 在 138,145 个 visited assignment nodes 中剪掉 122,362 个，剪枝率为 88.57%；
- Table III 中，两个算例的 first solution 等于 plan-graph best，另外两个仅高 1.83% 和 4.47%；
- 这些结果支持“先快速获得可执行计划，再持续改进”的 anytime 叙事。

4.2.2 严格审稿人会提出的质疑

1. **Warm start 的贡献没有被独立验证。**当前只有 warm-started BnB，没有“same BnB without warm start”。因此“warm start causes early pruning”仍是解释，不是消融证据。
2. **Fig. 2 的 upper-bound 曲线看起来频繁上升。**如果红线表示 incumbent upper bound，它理论上应单调不增。若它实际上是每个节点的 rollout cost，图例必须改名。
3. **“optimal value”用词过强。**Table III 显示 plan-graph best 并不总是等于 reference/global optimum。

4. **Table III 混合了两类问题。** GBA before/after 和 plan graph before/after 是结构压缩证据，应移动到组件消融；first/PG-best/reference cost 才属于质量分析。

若红线是全局 incumbent，应重新绘制为非增曲线 $J^{\star}(k)$；若红线是各扩展节点的局部 rollout cost，应改为“candidate rollout cost”。建议最终只画两条主曲线：incumbent makespan 和 minimum frontier lower bound，并用垂线标出 first feasible plan 和 best incumbent 的获得时刻。

4.2.3 推荐的新小节逻辑

可直接替换的英文结果段

The on-the-fly construction supplies an executable incumbent before the assignment search is exhausted. In the representative instance in Fig. X, the first plan has a makespan of 47.21*. The warm-started BnB search reduces this value to* 43.32*, corresponding to an* 8.24% *improvement, and obtains the best incumbent at* 2.3679 *s. Among the 138,145 visited assignment nodes, 122,362 are pruned, yielding a pruning ratio of* 88.57%*. These results show that the first plan is not only immediately executable but also provides a useful incumbent for subsequent pruning. The same trend is observed across the four inspection missions in Table X. The first plan coincides with the best solution retained in the plan graph in two cases. In the remaining cases, its makespan is only* 1.83% *and* 4.47% *higher, respectively. Therefore, the proposed ordering tends to expose a high-quality accepting branch early. Table X also distinguishes this plan-graph result from the reference optimum. The observed differences indicate that the BnB stage optimizes the assignments and paths retained in the generated plan graph; it should not be interpreted as a certificate of global optimality for the original LTL-MRTA problem.*

建议增加一个最小消融：

Method	$T_{\text{first prune}}$	nodes expanded to best	T_{best}
BnB with on-the-fly warm start	existing	existing	existing
BnB without warm start	to be added	to be added	to be added

该消融比继续增加新公式更重要，因为它直接验证 warm-start 这一算法设计。

4.3 当前 B：组件消融应从“两个变体”改为独立因素设计

4.3.1 当前问题

当前 Variant I 保留 GBA pruning 但关闭 on-the-fly planning；Variant II 同时关闭 GBA pruning 和 on-the-fly planning，再进行 post-processing。这个设计能证明 full method 优于两个较弱版本，但无法严格分离两个组件的独立贡献。

例如，Variant II 的时延增加究竟来自：

- 没有 early GBA pruning；
- 没有 on-the-fly planning；
- 两者同时关闭后的交互效应；

当前实验不能回答。

4.3.2 推荐的 2×2 设计

表 4: 推荐的组件消融配置

Configuration	Early GBA pruning	On-the-fly allocation	作用
Full	on	on	完整方法
No-OTF	on	off	单独测 on-the-fly allocation
No-Early-Pruning	off	on	单独测 early pruning
Offline/Post	off	off	传统“先构造完整 NBA 再规划”流程

如果暂时不能新增完整 2×2 数据，至少增加 No-Early-Pruning + On-the-Fly 这一项，否则不要写“Tables I and II isolate the two components independently”。

4.3.3 现有数据可以支持的结论

- 在相同 early-pruned GBA 下，on-the-fly planning 相对 Variant I 将 first-solution time 缩短约 18.1%–42.2%；
- 与先构造未剪枝 NBA 再 post-process 的 Variant II 相比，现有方法在 $m = 10$–15 时将 transition 数减少约 40.0%–60.0%；
- 对应的 first-solution speedup 为约 $1.22\times$–$2.23\times$；
- 结论应表述为“early elimination prevents unnecessary transitions from being generated and propagated”，而不是笼统的“planning-aware pruning improves efficiency”。

推荐的英文结果段

Table X evaluates the two mechanisms that distinguish the proposed construction from the conventional offline pipeline. When GBA-level pruning is fixed, enabling on-the-fly allocation reduces T_{first} *for all tested formula lengths because planning begins as soon as candidate NBA transitions are generated. The benefit becomes larger for longer specifications, for which waiting for the complete NBA accounts for a substantial fraction of the total latency.*

Early GBA pruning provides a complementary benefit. Compared with constructing the unpruned NBA and applying the same rules afterward, the proposed method avoids generating 40.0%–60.0% *of the NBA transitions in the tested instances. Post-processing can recover the same final automaton size, but it cannot recover the time and memory already spent on the removed transitions. Together, early pruning and on-the-fly allocation reduce both the amount of intermediate structure and the time before an executable inspection plan becomes available.*

4.4 当前 D 与 E：Scalability 和 Baseline Comparison 应融合

4.4.1 当前 Scalability 设计存在混杂变量

当前 D 中，n 既表示机器人总数，又进入任务需求：

$$c(a_1) = (1, n, l_2), \quad c(a_2) = (1, n, l_5), \quad c(a_3) = (1, \lfloor n/3 \rfloor, l_4).$$

因此，当 n 从 5 增加到 100 时，robot pool 和 task demand 同时变化。该实验不能被严格称为“scalability with respect to the number of robots”。

将规模实验拆成两条轴，但仍放在同一个小节：

1. **Team-size scaling**: 固定 n_i，只增加可用机器人数量；
2. **Collaboration-demand scaling**: 固定机器人总数，只增加 n_i 或需求比例。

前者验证算法对更大 robot pool 的适应性，后者验证同步协作任务的组合复杂度。当前 Table IV 可保留为 coupled stress test，但不能作为纯 team-size scalability 的唯一证据。

4.4.2 Table V 太大且讨论不完整

Table V 包含 10 个 formula/team groups、3 种方法和 6 个指标，占据接近整页。主要问题不是数据多，而是：

- 主文表格过密，读者难以看出趋势；
- φ_2、φ_3、φ_4 的多个 team size 可以画成 log-scale 曲线；
- 文中只强调 proposed 更快，但没有讨论 φ_1 中 MILP 的最终代价 32.85 明显低于 proposed 的 51.48；
- ∞ 应替换为 TO，并说明 timeout；
- 需要明确所有方法是否使用相同地图、初始状态、低层距离、随机种子和终止条件。

推荐将主文改为：

1. 一张图：T_{first} 随总机器人数量变化，纵轴对数；
2. 一张图：J_{best} 或 fixed-budget gap；
3. 一张小表：只保留 simple/medium/complex 三个代表性点；
4. 完整 Table V 移至 appendix 或 supplementary material。

4.4.3 基于现有数据的合理结论

- $\varphi_2, 5 \times 20$：proposed 的 $T_{\text{first}} = 1.1497$ s，分别比 poset 和 MILP 快约 $78.4\times$ 和 $58.3\times$；
- $\varphi_4, 5 \times 20$：对应 speedup 约为 $152.1\times$ 和 $272.2\times$；
- 对 φ_2–φ_4，三种方法在当前表中达到相同最终代价，说明主要优势是更早获得可执行方案；
- 对 φ_1，MILP 得到更低最终代价，表明 proposed 的主要优势不是在所有任务上都获得最低代价，而是在复杂任务上显著降低 first-solution latency。

推荐的英文讨论段

The principal advantage of the proposed method is the latency of the first executable plan. For φ_2 with a 5×20 team, the proposed method returns a plan in 1.1497 s, whereas the poset- and MILP-based methods require 90.137 s and 67.0037 s, respectively. For the more complex φ_4 instance, the corresponding speedups are approximately $152.1\times$ and $272.2\times$. The three methods reach the same final cost for the tested φ_2–φ_4 instances, but the proposed method reaches that cost substantially earlier.

The result for φ_1 also clarifies the scope of this advantage: the MILP baseline obtains a lower final cost, while the proposed method returns its first plan much earlier. Therefore, the experiments support a latency–quality tradeoff *rather than uniform dominance in final objective value. This distinction is important for inspection missions in which execution must begin promptly and further optimization can continue after a valid plan is available.*

4.5 当前 F：Online Repair 需要增加直接对照

4.5.1 优点

该实验与论文实际应用最接近：机器人在执行中失效，原 LTL mission 不变，但未完成 assignment 必须从当前状态重新分配。使用 Gantt chart 展示 failure time 和 reassignment 是合理的。

4.5.2 需要修复的问题

- 只有 proposed workflow 的绝对时间，没有"从当前状态重新构造 NBA/plan graph 并规划"的对照；
- 第二次故障的 first repair time 更短，不能解释为"更严重故障也更快"，因为两次事件的 residual mission 不同；
- agent/robot 混用；应统一为 robot；
- "agent 0 breaks down"口语化，应改为"robot r_0 becomes unavailable"；
- Fig. 3 caption "Gantt graph of planning"信息不足，应标出故障时刻、被取消任务和重新分配任务；
- 当前只验证单条执行轨迹，建议增加多个 failure time、failed type 和 failed count 的统计结果。

推荐的英文结果段

We next evaluate online repair in an industrial inspection mission. Robot r_0 becomes unavailable at $t = 15$ s, and robot r_{20} becomes unavailable at $t = 30$ s. After each event, assignments involving the unavailable robot are invalidated, while completed subtasks and unaffected assignments are retained. The planner re-roots the remaining search at the confirmed execution state and reallocates only the unfinished subtasks.
The first repaired plans are obtained in 45.738 ms and 1.652 ms, and the corresponding best repaired plans are obtained in 69.032 ms and 33.599 ms. These values are case-specific because the two events occur at different mission stages and leave different residual tasks. Fig. X shows that no task after a failure is assigned to the unavailable robots and that affected inspection and maintenance actions are transferred to compatible robots. The repaired plan preserves the original precedence and collaboration constraints and completes the mission at approximately 62 s.

为突出创新点，必须增加一列或一张小表：

Method	$T_{\text{repair,first}}$	$T_{\text{repair,best}}$	reused plan-graph nodes
Retained-graph repair	existing	existing	report
Full replanning from current state	add	add	0

这比再增加第三次故障更有说服力，因为它直接验证"复用保留 plan graph"这一机制。

4.6 空的 Physical Experiments 是投稿风险

当前 Section XII 只有标题，没有内容，但摘要写有“hardware experiments”，结论也写有“Numerical and physical experiments validate...”。这是严格审稿中会被立即指出的问题。

1. 补充物理巡检实验：至少说明平台、传感器、地图、任务公式、机器人失效触发方式、planning latency 和执行结果；
2. 在物理数据完成前，从 Abstract、Introduction、Conclusion 和 Section organization 中删除 physical/hardware experiment 的声明，并删除空标题。

5 现有结果的定量解读与可支持结论

5.1 Early solution 与 BnB improvement

表 5: 根据现有数据可直接报告的关键量

证据	数值	可支持的结论
代表性 BnB 改进	$47.21 \rightarrow 43.32$，降低 8.24%	first plan 是有效 incumbent，后续搜索可以继续改进
BnB 剪枝率	$122{,}362/138{,}145 = 88.57\%$	初始 incumbent 对 assignment search 有实际剪枝价值
First vs plan-graph best	gap 为 $0, 0, 1.83\%, 4.47\%$	residual-obligation ordering 通常较早发现高质量 branch
Plan-graph best vs reference optimum	gap 为 $21.97\%, 9.35\%, 0, 15.23\%$	不应宣称原问题全局最优；应限定为 retained graph 内优化

5.2 Early pruning 与 on-the-fly construction

表 6: 现有消融数据的合理解释范围

比较	现有范围	合理结论
Full vs Variant I	$1.22\times$–$1.73\times$ first-solution speedup	在相同 GBA pruning 下，on-the-fly allocation 减少等待完整 NBA 的时间
Full vs Variant II	$1.22\times$–$2.23\times$ speedup	integrated pipeline 优于 offline construction + post-processing
Transition reduction	40.0%–60.0%	early pruning 避免生成随后会被删除的 NBA transitions

5.3 Baseline comparison 的正确叙事

实验最强证据是 first-solution latency，而不是“所有情况下最终代价更好”。因此建议将创新点对应到以下三句：

1. The proposed method begins assignment before the NBA is complete.
2. It eliminates infeasible or decomposable GBA transitions before they are propagated to the NBA stage.
3. The resulting executable plan warm-starts a subsequent assignment optimizer.

对应实验结论应写成：

- complex inspection missions 下，T_{first} 显著缩短；
- first solution 通常接近 plan-graph best；
- final cost 并非对所有 simple instances 都优于 MILP；
- 当完整枚举 timeout 时，框架仍能早期返回可执行计划，体现 anytime 性质。

6 推荐的实验矩阵与统计协议

表 7: 推荐的 research-question-driven 实验矩阵

RQ	主要变量	对照	指标
RQ1	early pruning on/off；on-the-fly on/off；formula length m	2×2 configurations	T_{first}，GBA/NBA transitions，plan-graph nodes/edges，peak memory
RQ2	search budget；warm start on/off；mission class	no-warm-start BnB	J_{first}，best incumbent，gap vs PG-best，nodes expanded，pruning ratio
RQ3	fixed-demand team size；fixed-team collaboration demand；formula class	poset、MILP	T_{first}，T_{best}，TO rate，cost gap，speedup
RQ4	failure time、failed type、failed count	full replanning from current state	repair latency，success rate，reused nodes，reassigned tasks，mission makespan

6.1 统计规范

1. 所有随机初始状态和随机障碍实验使用相同 seeds；
2. 每个配置至少重复 20–30 次；当前已使用 25 次的部分可统一推广；
3. 时间指标建议报告 median [IQR] 或 mean $\pm$ standard deviation，并给出 95% confidence interval；
4. 对 timeout 数据，不要用 ∞ 参与均值，应报告 TO count / success rate；
5. 基线必须使用相同机器、公式、地图、机器人初始状态、距离缓存和停止条件；
6. 对 first-solution latency，起始时刻必须统一为“提交 LTL formula 后”；
7. “global optimum”只有在原问题被完整穷举或有独立精确求解器证明时才能使用，否则用 reference optimum；
8. 除运行时间外，建议报告 peak memory，因为 early pruning 的直接收益还包括中间结构减少。

7 可直接替换的完整英文 Section XI 框架

下面给出一个基于现有结果、但避免过度声明的重写版本。需要新增的数据以括号说明，不应原样留在最终论文中。

XI. Numerical Experiments

This section evaluates the proposed method in a large industrial inspection scenario. The numerical study addresses four questions: 1) whether GBA-level pruning and on-the-fly allocation reduce the structure generated before the first executable plan; 2) whether the first plan provides a useful solution and warm start for subsequent optimization; 3) how the method scales with mission and team size compared with representative baselines; and 4) whether the retained plan graph supports rapid repair after robots become unavailable.

All methods were implemented in Cython 0.29.24 and executed on a workstation equipped with a 3.40-GHz Intel Core i7-11390H CPU and 16 GB RAM. The workspace is a $40\,\mathrm{m} \times 40\,\mathrm{m}$ *industrial facility containing eight inspection regions and randomly generated obstacles. The heterogeneous team includes mobile visual-inspection robots, thermal-inspection robots, environmental-sensing robots, maintenance robots, and support robots. An action proposition specifies the required robot type, team size, and inspection region through* $c(a) = (\tau_a, n_a, l_a)$*. Unless otherwise stated, all methods use identical initial states and random seeds. We report the first-solution time* T_{first}*, the time to the best incumbent* T_{best}*, and the termination time* T_{search}*. A run exceeding the prescribed time limit is marked TO.*

A. Ablation of Integrated Automaton Construction

We first isolate the mechanisms that reduce the delay before an executable plan is available. The parametric mission $\varphi_m = \bigwedge_{i=1}^{m} \Diamond a_i$ *requires inspections in an increasing number of regions. The full method is compared with configurations that disable on-the-fly allocation, early GBA pruning, or both. All configurations use the same team and initial states.*

When GBA-level pruning is fixed, on-the-fly allocation consistently reduces T_{first} *because the plan graph is extended during NBA generation rather than after the NBA is complete. Early GBA pruning provides a complementary reduction in intermediate structure. Compared with generating the unpruned NBA and applying the same rules afterward, the proposed method avoids* 40.0%*–*60.0% *of the NBA transitions in the tested instances. Although post-processing reaches the same final automaton size, it cannot recover the time and memory spent on transitions that have already been generated. These results show that the two components act at different stages: early pruning reduces the transitions propagated to the NBA, whereas on-the-fly allocation reduces the waiting time before planning begins.*

需要新增： No-Early-Pruning + On-the-Fly 配置，或删除“independently isolates”类表述。

B. First-Solution Quality and Warm-Started Optimization

We next evaluate the anytime behavior of the proposed framework. In the representative instance in Fig. X, the on-the-fly plan graph returns an executable plan with a makespan of 47.21*. The warm-started BnB search reduces this value to* 43.32*, an improvement of* 8.24%*, and obtains the best incumbent at* 2.3679 *s. The search prunes 122,362 of the 138,145 visited assignment nodes, corresponding to a pruning ratio of* 88.57%*.*

Table X reports the quality of the first plan across four inspection missions. The first plan coincides with the best solution retained in the plan graph in two cases and is only 1.83% *and* 4.47% *higher in the other two. Thus, the residual-obligation ordering generally exposes a high-quality accepting branch early. The table also reports a reference optimum obtained by the independent exact procedure used in the experiment.*

The differences between the plan-graph best and the reference optimum clarify that the BnB stage optimizes the paths and assignments retained in the generated plan graph; it does not certify global optimality for the original problem when the graph construction has removed alternatives.

建议新增： same BnB without warm start，以直接证明 warm-start 的节点剪枝收益。

C. Scalability and Comparison With Baselines

To isolate scalability with respect to team size, the action requirements are fixed while the number of available robots is varied. A second experiment fixes the team size and increases the number of robots required by synchronized inspection actions. This separation distinguishes the cost of evaluating a larger robot pool from the combinatorial cost of stronger collaboration requirements.

当前数据若不重跑，应将上一句改为：

The current experiment jointly increases the team size and the collaboration requirements and is therefore interpreted as a coupled stress test rather than a pure team-size scalability study.

The proposed method is also compared with poset- and MILP-based approaches under common mission formulas and team configurations. Its main advantage is first-solution latency. For φ_2 *with a* 5×20 *team, the proposed method returns an executable plan in* 1.1497 *s, compared with* 90.137 *s and* 67.0037 *s for the poset and MILP baselines. For* φ_4 *with the same team size, the corresponding speedups are approximately* $152.1\times$ *and* $272.2\times$*. The methods reach the same final cost in the tested* φ_2*–*φ_4 *instances, but the proposed method reaches this cost much earlier. For* φ_1*, the MILP baseline obtains a lower final cost, showing that the proposed method provides a latency–quality tradeoff rather than uniform dominance in objective value.*

For large teams, the BnB search may not exhaust the assignment space within the time limit, while the on-the-fly stage still returns an executable plan promptly. This behavior is consistent with the complexity analysis: greedy extension grows moderately with the number of compatible robots, whereas exhaustive assignment enumeration remains combinatorial.

D. Online Repair Under Robot Unavailability

Finally, we evaluate online repair during an inspection mission. Robot r_0 *becomes unavailable at* $t = 15$ *s, and robot* r_{20} *becomes unavailable at* $t = 30$ *s. After each event, assignments involving the unavailable robot are removed, while completed subtasks and unaffected assignments are retained. The remaining search is re-rooted at the confirmed execution state.*

The first repaired plans are obtained in 45.738 *ms and* 1.652 *ms, and the best repaired plans are obtained in* 69.032 *ms and* 33.599 *ms. These times correspond to different residual missions and should not be interpreted as a monotonic function of the number of failures. Fig. X shows that unavailable robots receive no subsequent tasks and that affected inspection actions are reassigned to compatible robots. The repaired execution preserves the original precedence and synchronization constraints and completes the mission at approximately* 62 *s.*

需要新增： retained-graph repair 与 full replanning from the current state 的直接对照。

8 表格和图片的具体修改方案

8.1 Table I 和 Table II 合并

建议合并为一张双面板表：

- Panel A: on-the-fly on/off，在 early pruning 固定时比较 T_{first}；
- Panel B: early pruning 与 post-pruning，比较 NBA states/transitions、peak memory 和 T_{first}；
- 增加 No-Early-Pruning + On-the-Fly 后，可以形成完整 2×2。

推荐 caption：

Ablation of GBA-level pruning and on-the-fly task allocation. Panel A isolates the effect of starting allocation during NBA generation. Panel B compares early pruning with constructing the unpruned NBA and applying the same rules afterward.

8.2 Table III 拆分

- 将 First Solution、Plan-Graph Best、Reference Optimum 留在 anytime quality 表；
- 增加两列：First-to-PG gap 和 PG-to-reference gap；
- 将 GBA Before/After、Plan Graph Before/After 移至组件消融表；
- 将 Global Optimum 改为 Reference Optimum，除非有严格全局证明。

8.3 Table IV 和 Table V 改为图表组合

- Figure X(a): T_{first} vs total robots，log scale；
- Figure X(b): T_{best} 或 fixed-budget cost gap；
- 小表只保留 simple/medium/complex 三个代表性配置；
- 完整 Table V 放 appendix；
- 所有 ∞ 改为 TO，并在 caption 中给出 timeout。

8.4 Fig. 2

推荐 caption：

Evolution of the incumbent makespan and the minimum lower bound in the BnB frontier. The vertical markers indicate the first feasible plan and the best incumbent.

注意：incumbent makespan 必须为非增曲线。若当前红线不是 incumbent，应重新命名。

8.5 Fig. 3

推荐 caption：

Execution schedule before and after the removal of robots r_0 and r_{20}. Vertical dashed lines indicate failure times; hatched segments denote inspection tasks reassigned after each event.

图中建议增加：

- 两条竖直 failure markers；
- failed robot 的任务取消标记；
- reassigned task 的 hatch 或边框；
- action proposition 到巡检操作的图例；
- 统一使用 robot，而不是 agent。

9 句子与术语的直接修改示例

当前表达	推荐表达
This section evaluates the proposed framework through six case studies...	This section evaluates the proposed method through four research questions.
This case study examines whether...	We first evaluate... / We next examine... / Finally, we test...
This is mainly because that the warm start provides...	The warm-start incumbent enables pruning from the first BnB iterations.
As shown in Tab. V, the results show that...	Table V shows that...
planning-aware automaton construction	integrated automaton construction and task allocation
Global Optimum	Reference Optimum, unless global optimality is independently certified
Best solution	best incumbent / plan-graph best / reference optimum, according to scope
$T_{\text{search}} = \infty$	TO under a stated time limit
agent 0 breaks down	robot r_0 becomes unavailable
Gantt graph of planning	Gantt chart of the repaired execution schedule
large-scale factory scenarios	large industrial inspection and maintenance missions
The results demonstrate the anytime nature...	The early feasible plan remains available when exhaustive assignment search does not terminate within the time limit.

10 投稿前修改优先级

P0：必须完成

1. 修复 Scalability 中 team size 与 task demand 同时变化的问题，或更名为 coupled stress test；
2. 增加 No-Early-Pruning + On-the-Fly，避免消融归因不清；
3. 将 Global Optimum 改为 Reference Optimum，并同步收紧理论声明；
4. 检查 Fig. 2 的 upper bound 定义和单调性；
5. 在线修复增加 full replanning baseline；

6. 补充 Physical Experiments，或删除所有相关声明。

P1：强烈建议

1. 将六组 case studies 融合为四组 research questions；
2. 用统一巡检场景解释区域、机器人类型和 action propositions；
3. 统一重复次数、随机种子、统计量和 timeout；
4. 将大 Table V 转为曲线 + 小表，完整数据移至 appendix；
5. 增加 warm-start vs no-warm-start 的直接消融。

P2：语言与版式

1. 删除反复出现的“This case study...”和“These results indicate...”；
2. 统一 robot/agent、first solution/initial plan、best incumbent/plan-graph best；
3. 修复 Table ??、Sec.XI-A、Tab.V 等交叉引用和空格问题；
4. 将 action requirement 的长列表集中到一张表，不在多个小节重复。

11 最终评价

当前实验数量本身并不过量，且已有数据能够支撑论文最重要的 practical message：**对于复杂的异构机器人巡检任务，方法可以在完整自动机构造和组合分配搜索结束之前返回可执行计划，并允许后续继续优化和失效后修复。**

但当前写法把这一主线拆成了六个平行算例，导致创新点分散、统计口径不统一、部分因果归因不足。按照本报告建议重组后，实验会形成清晰的四段证据链：

1. 为什么早期剪枝和同步构造有效；
2. 为什么首解既快又有实用质量；
3. 在规模增大和外部方法比较中优势体现在哪里；
4. 在巡检机器人失效后为何能够快速恢复任务。

审稿意见仍建议为 **Major Revision**，但主要修改集中在实验设计归因、叙事组织和少量关键对照，不需要推翻已有实验体系。